

PW EARNERS
DSA in C++

Daily Practice Problems (DPP) — 7
Topic: 2D Arrays and 2D Vectors

Multiple Choice Questions (Predict the Output)

For Q1 to Q7: Select the exact output of each program snippet. Assume all required headers like `<iostream>` and `<vector>` are included, and `using namespace std;` is present.

Q1.

```
1 int main(){
2     int arr[2][3] = {1, 2, 3, 4, 5};
3     cout << arr[1][1] << " " << arr[1][2];
4 }
```

A. 2 3

C. 4 5

B. 5 0

D. Garbage Value

Q2.

```
1 int main(){
2     vector<vector<int>>> v(3, vector<int>(2, 7));
3     cout << v[2][1];
4 }
```

A. 0

C. 2

B. 3

D. 7

Q3.

```
1 void solve(int arr[][]) {
2     cout << arr[0][0];
3 }
4 int main(){
5     int arr[2][2] = {{1, 2}, {3, 4}};
6     solve(arr);
7 }
```

A. 1

C. Segmentation Fault

B. Compilation Error

D. Garbage Value

Q4.

```

1 void modify(vector<vector<int>> v) {
2     v[0][0] = 99;
3 }
4 int main(){
5     vector<vector<int>> vec = {{1, 2}, {3, 4}};
6     modify(vec);
7     cout << vec[0][0];
8 }

```

- A. 99 B. 1 C. Compilation Error D. 0

Q5.

```

1 int main(){
2     int arr[3][3] = {{1, 2, 3}, {4, 5, 6}, {7, 8, 9}};
3     cout << *((arr + 1) + 2);
4 }

```

- A. 2 B. 5 C. 6 D. 8

Q6. Crazy Question

```

1 int main(){
2     int arr[2][2] = {{10, 20}, {30, 40}};
3     int (*ptr)[2] = arr;
4     cout << **(&ptr + 1) << " " << **(&ptr + 1);
5 }

```

- A. 30 20 C. 30 40
B. 20 30 D. Compilation Error

Q7. Crazy Question

```

1 int main(){
2     vector<vector<int>> v;
3     v.push_back({1, 1});
4     int& ref = v[0][0];
5
6     for(int i = 0; i < 100; i++) {
7         v.push_back({i, i});
8     }
9
10    ref = 100;
11    cout << v[0][0] << " " << v.size();
12 }

```

- A. 100 101 C. 100 100
B. 1 101 D. Undefined Behavior (Dangling Reference)

For Q8 to Q14: Write clean, optimal, and compilable C++ programs. Assume inputs are provided by the user. Do NOT use predefined STL algorithms like `sort()` or `reverse()` unless implicitly required by the logic.

Q8. Row-wise Strictly Increasing. Write a function that returns `true` if every row in a given $M \times N$ matrix is strictly increasing from left to right, otherwise return `false`.

Constraints: $1 \leq M, N \leq 10^3$, $-10^5 \leq arr[i][j] \leq 10^5$.

Sample Input: $M = 2, N = 3$, Matrix = $\{\{1, 5, 9\}, \{2, 3, 4\}\}$

Sample Output: `true`

Q9. Count Negatives. Given an $M \times N$ matrix which is sorted in non-increasing order both row-wise and column-wise, count the number of negative numbers. Find an optimal approach that runs in $O(M + N)$ time complexity.

Constraints: $1 \leq M, N \leq 10^4$, $-10^5 \leq arr[i][j] \leq 10^5$.

Sample Input: $\{\{4, 3, 2, -1\}, \{3, 2, 1, -1\}, \{1, 1, -1, -2\}, \{-1, -1, -2, -3\}\}$

Sample Output: 8

Q10. Toeplitz Matrix. A matrix is a Toeplitz matrix if every diagonal from top-left to bottom-right has the same elements. Write a function to check if a given $M \times N$ matrix is a Toeplitz Matrix. Return `true` or `false`.

Constraints: $1 \leq M, N \leq 10^3$.

Sample Input: $\{\{1, 2, 3, 4\}, \{5, 1, 2, 3\}, \{9, 5, 1, 2\}\}$

Sample Output: `true`

Q11. Rotate Matrix by 180 Degrees. Given an $N \times N$ square matrix, rotate it by 180 degrees in-place. You must not use any extra 2D array for the final result.

Constraints: $1 \leq N \leq 10^3$.

Sample Input: $\{\{1, 2\}, \{3, 4\}\}$

Sample Output: $\{\{4, 3\}, \{2, 1\}\}$

Q12. Saddle Point. A saddle point in a matrix is an element which is the **minimum** element in its row and the **maximum** element in its column. Find and print the saddle point if it exists, otherwise print -1. Assume all elements are distinct.

Constraints: $1 \leq M, N \leq 500$.

Sample Input: $\{\{1, 2, 3\}, \{4, 5, 6\}, \{7, 8, 9\}\}$

Sample Output: 7 (7 is the minimum in the 3rd row, and the maximum in the 1st column)

Q13. Shift 2D Grid. Given a 2D grid of size $M \times N$ and an integer K . You need to shift the grid K times. A single shift operation moves:

- Element at `grid[i][j]` to `grid[i][j + 1]`
- Element at `grid[i][N - 1]` to `grid[i + 1][0]`
- Element at `grid[M - 1][N - 1]` to `grid[0][0]`

Constraints: $1 \leq M, N \leq 50, 0 \leq K \leq 100$.

Q14. Anti-Diagonal Traversal. Given an $M \times N$ matrix, return an array of all its elements in anti-diagonal order (starting from the top-left to the bottom-right).

Constraints: $1 \leq M, N \leq 1000$.

Sample Input: $\{\{1, 2, 3\}, \{4, 5, 6\}, \{7, 8, 9\}\}$

Sample Output: 1, 2, 4, 3, 5, 7, 6, 8, 9

All the best! Keep Coding ♡